*Session 2 · Tuesday 1 September 2026 · Start wherever the scaffolding stops helping you. Nobody is watching where you start.*

# Four estimates

Four versions of one problem, with **less of my working shown each time**. Item 1 is done for you; item 4 is not. **Start wherever the scaffolding stops helping you** — nobody has to announce which. The **bold labels** say what each step *accomplishes*; that is the part worth carrying forward. At each ▶ **Why does that step follow?**, answer in writing before going on.

| | | | |
|---|---|---|---|
| AVOGADRO | $N_A = 6.0 \times 10^{23}\ \mathrm{mol}^{-1}$ | *E. COLI* | $1\ \mathrm{fL} = 10^{-15}\ \mathrm{L}$, ~1 µm across |
| HELA CELL | ~20 µm across | GENOME | $4.6 \times 10^{6}$ bp |
| GFP IN CYTOPLASM | $D = 7.7\ \mu\mathrm{m}^2/\mathrm{s}$ *(Elowitz 1999)* | DIVISION | 20–30 min, rich medium |

## 1 · Fully worked

> *A repressor is present at **10 nM** in E. coli. How many molecules is that?*

**Convert the concentration to moles in this volume.**

$$n = cV = (10 \times 10^{-9}\ \mathrm{mol/L})(10^{-15}\ \mathrm{L}) = 10^{-23}\ \mathrm{mol}$$

**Convert moles to a count.**

$$N = nN_A = (10^{-23})(6.0 \times 10^{23}) = 6\ \mathrm{molecules}$$

**State the answer at the precision you actually have.**

About **6 molecules per cell** — call it "of order ten". Not 6.02: the cell volume is known to maybe a factor of two, so the second digit is fiction.

> ***Worth carrying:*** $1\ \mathrm{nM} \approx 1$ *molecule per E. coli. Every other copy number in this course is that number times something.*

▶ **Why does that step follow?** Why is it legitimate to report one significant figure when the arithmetic gave you three?

## 2 · The last step is yours

> *How long does a protein take to diffuse across E. coli?*

**State the convention before the number.**

For one-dimensional displacement, $\langle x^2 \rangle = 2Dt$, so the time to cover a distance $L$ is $t \sim L^2/2D$. *(In three dimensions it is $L^2/6D$. The factor of three does not matter here. Not saying which one you used does.)*

**Put in the measured numbers.**

$$L = 1\ \mu\mathrm{m}, \qquad D = 7.7\ \mu\mathrm{m}^2/\mathrm{s}$$

**Now you: evaluate it, and say it in words.**

---

---

▶ **Why does that step follow?** Why use the *measured* $D$ in cytoplasm rather than the value for the same protein in water — and which way would the answer go if you used the wrong one?

---

## 3 · The last two steps are yours

> *The same protein, in a HeLa cell instead. How long, and how many times longer than in E. coli?*

**State the convention.** Same as item 2 — $t \sim L^2/2D$, one-dimensional.

**Now you: everything else.** Take $D$ to be the same, and say at the end whether that assumption makes your answer too big or too small.

▶ **Why does that step follow?** The cell is 20 times wider. Why is the answer not 20 times longer?

## 4 · Bare problem

> *A repressor must find **one specific operator site** in the E. coli genome. Suppose it searches by diffusing through the cytoplasm until it happens to collide with the right sequence.*
>
> *Estimate how long that takes, and compare it to the cell cycle.*

You have everything you need. State your assumptions as you go, and at the end say which one you are least confident in.

▶ **The real question.** The measured time is of order *minutes*. Whatever you got is probably two orders of magnitude away from that — and **which side** you landed on depends entirely on the assumption you made.

Your estimate is not wrong because you made an arithmetic error. It is wrong because the *mechanism* is not what the problem said it was.

**What could the protein be doing instead?**

---

*An estimate that disagrees with a measurement is not a failure. It is the only reliable way anyone has ever found a mechanism they were not looking for.*